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Request for Extension of MSc

Dear Committee Members,

I am writing to provide an explanation for the delay in completing my master's research and to request an extension of the study period to 31 March 2027. The accompanying Gantt chart shows the planned timeline for the remaining research and submission before the end of March 2027.

My research focuses on developing a synthetic aperture radar (SAR) based training dataset and machine learning model for detecting floating marine plastics in the South African exclusive economic zone. The original research plan was developed around obtaining an existing SAR-labelled dataset from a research projected called Marine Litter Signatures in SAR Images (MIREA). This dataset is the only of its kind and was intended to provide an initial foundation for training and evaluating a machine learning model, which could then be refined using South African satellite imagery. Therefore, a large portion of my first year was dedicated to developing a methodology on the assumption that suitable training data would be available.

Despite numerous attempts to obtain the dataset from the original research team, it ultimately became clear that the required data would not be available. This introduced additional complexity, as the research now required the development of a labelled SAR dataset before model training and evaluation could begin.

Creating this dataset involves identifying marine debris in optical imagery, matching these observations with corresponding Sentinel-1 SAR imagery, modelling the drift of the debris over the temporal gap between the sensors and then labelling the debris in SAR to create a training dataset.

Although this change has extended the research timeline, it has also strengthened the potential contribution of the dissertation by addressing the lack of labelled marine debris in SAR.

The revised plan is to complete data labelling and patch preparation by the end of November 2026, followed by model training during December. Model evaluation and validation will take place in January 2027, with the remaining dissertation writing, including the methodology, results, discussion and conclusion, scheduled for February and March.

Thank you for considering my application. I am confident that the extension will allow me to complete the remaining research and make a meaningful contribution to marine plastic detection with SAR.

Kind regards,

Joshua Pretorius

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